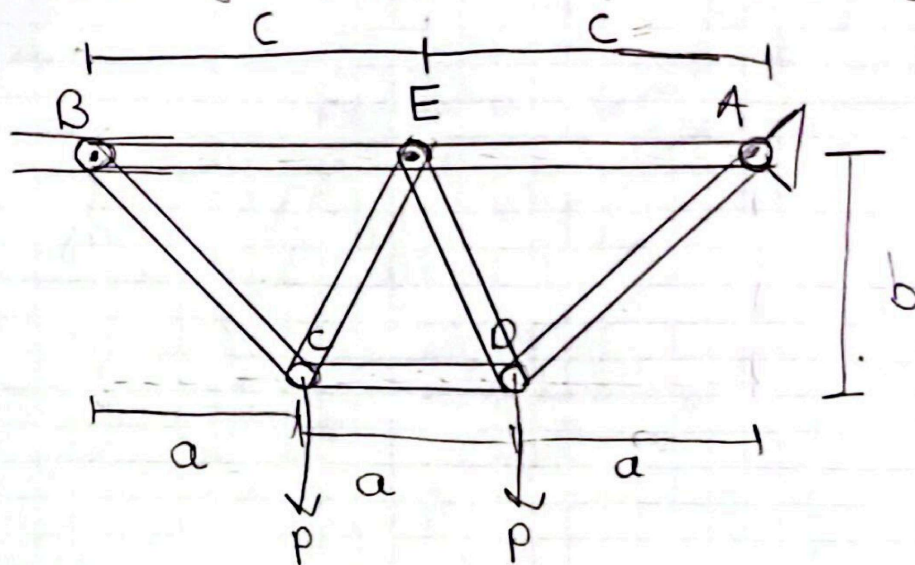


## A2. Design with Basic stresses



Design

- used triangularization

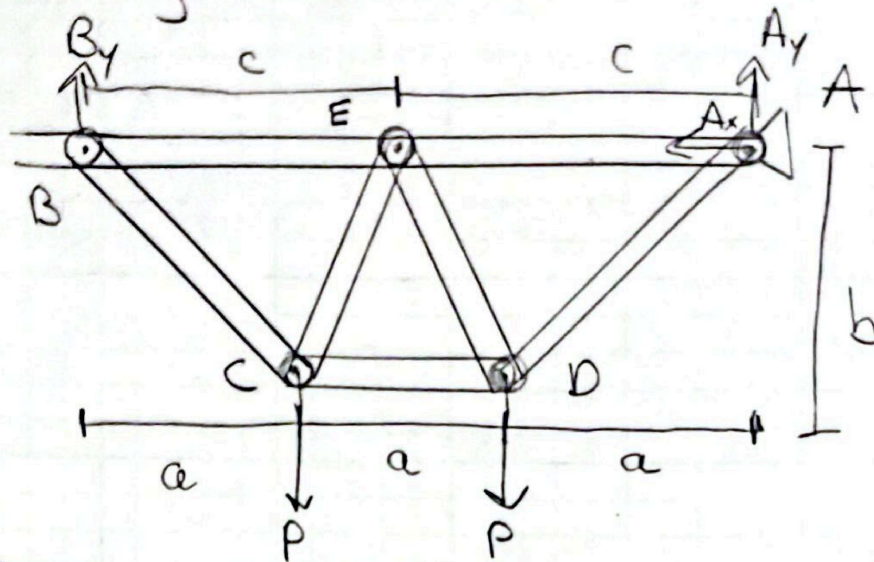
$$2(\# \text{ joints}) = 3 + \text{members}$$

$$2(5) = 3 + 7$$

E was added in the center between B and A

$$c = \frac{3a}{2}$$

Solving for unknowns



Given S

$$a = .4\text{m}, b = .3\text{m}, c = .6\text{m}, P = 27.1\text{kN}$$

B is a roller, A is a pin

find:  $A_x, A_y, B_y$

$$\text{Equilibrium} \Rightarrow \sum F_x = 0, \sum F_y = 0, \sum M = 0$$

$$\sum F_x = -A_x = 0$$

$$\sum F_y = B_y + A_y - 2(P) = 0$$

$$\sum M_B = -(a)P - 2(a)P + 2cA_y = 0$$

$$2cA_y = 3aP$$

$$A_y = \frac{3aP}{2c} = \frac{3(.4)(27.1 \times 10^3)}{2(.6)}$$

$$A_y = 27100\text{N}$$



Using  $\Sigma F_y$  equation

$$B_y + A_y - 2(P) = 0$$

$$-B_y = A_y - 2(P) = 2700 - 2(27.1 \times 10^3)$$

$$-B_y = -27100$$

$$B_y = 27100 \text{ N}$$

$$\begin{aligned}
 BE &= 36200 \text{ N C}, BC = 45200 \text{ N T}, CE = 24.1 \text{ N C} \\
 CD &= 36200 \text{ N T}, DE = 24.1 \text{ N T}, EA = 36200 \text{ N C} \\
 DA &= 45300 \text{ N T},
 \end{aligned}$$

largest Internal force?

$$DA = 45300 \text{ N}$$

Find the required cross-sectional area

Given: Safety factor = 3.5,  $DA = 45300 \text{ N}$   
 Yield Strength of A500 Steel = 230 MPa

$$\sigma = \frac{F}{A}, \quad \frac{S_y}{SF} > \sigma$$

$$\frac{S_y}{SF} > \frac{F}{A} \Rightarrow \frac{S_y}{SF} = \frac{F}{A_{\min}}$$

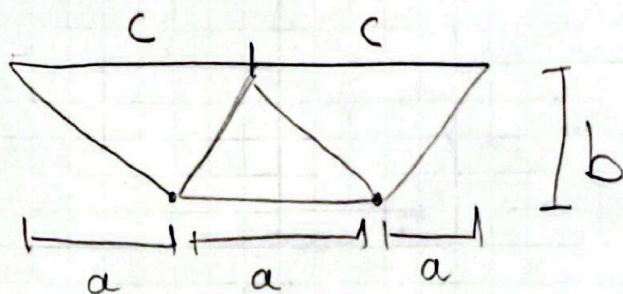
$$A_{\min} = \frac{DA \cdot SF}{S_y} = \frac{45300 \text{ N} \cdot 3.5}{230 \text{ MPa}}$$

$$A_{\min} = 689 \text{ mm}^2$$



Find Approximate weight

$$A = 689 \text{ mm}^2, \rho = 7,800 \text{ kg/m}^3$$



We know the crosssectional area. Now we have to find the length of the entire truss and multiply it with the crosssectional area to get the approximate volume. Then we can use this calculated volume value and given density to find the mass.

$$2(c) + a + \sqrt{a^2 + b^2} (2) + 2(\sqrt{a^2 + b^2})$$

$$2(1.6) + .4 + \sqrt{.4^2 + .3^2} (2) + 2(\sqrt{.2^2 + .3^2})$$

$$L = 3.32 \text{ m}$$

$$V = L \cdot W \cdot h$$

$$V = L \cdot A$$

$$V = 3.32 \cdot 689 \times 10^{-6} = 0.002287 \text{ m}^3$$

$$\rho = 7,800 \text{ kg/m}^3$$

$$\rho = \frac{m}{V} \quad m = \rho V$$

$$7800 \frac{\text{kg}}{\text{m}^3} \cdot 0.002287 \text{ m}^3 = 17.839 \text{ kg}$$

$$W = M \cdot g$$

$$W = 17.84 \text{ kg} \cdot 9.81 \frac{\text{m}}{\text{s}^2} = 175 \text{ N}$$

$$\tau = \frac{V}{A}$$

$$\frac{SS_y}{SF_s} > \tau$$

$$\frac{SS_y}{SF_s} > \frac{V}{A}$$

$$\frac{SS_y}{SF_s} = \frac{V}{A_{min}}$$

$$A_{min} = \frac{V SF_s}{SS_y} = \frac{45300N \cdot 4}{1172.1 \text{ MPa}}$$

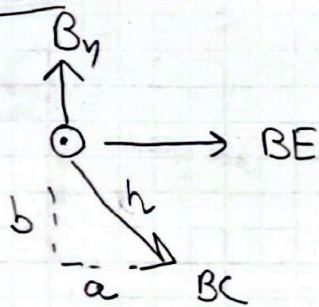
$$A_{min} = 155 \text{ mm}^2$$



FBD of each joint

Joint B

FBD



Equations of equilibrium

$$\sum F_x = BE + \frac{a}{h} BC = 0$$

$$\sum F_y = B_y - \frac{b}{h} BC = 0$$

using  $\sum F_y$

$$B_y = \frac{b}{h} BC$$

Angle of BC

$$\frac{b}{a} \quad \frac{h}{a} \quad b^2 + a^2 = h^2$$

$$\sqrt{b^2 + a^2} = h$$

$$h = \sqrt{.3^2 + .4^2} = .5$$

$$\frac{h}{b} B_y = BC$$

$$\frac{.5}{.3} (27100) = BC$$

$$BC = 45200 \text{ N/T}$$

$\sum F_x$

$$BE + \frac{a}{h} BC = 0$$

$$BE = -\frac{a}{h} BC$$

$$BE = -\frac{.4}{.5} (45200)$$

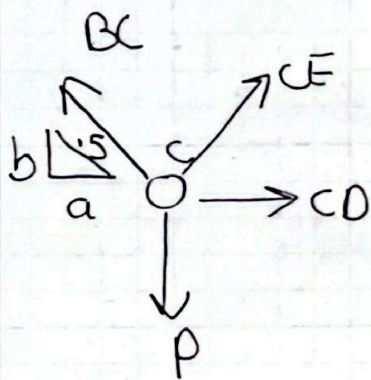
$$BE = -36160 \text{ N} \Rightarrow$$

$$BE = 36200 \text{ N C}$$

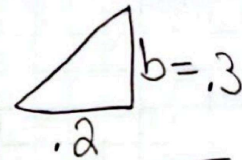
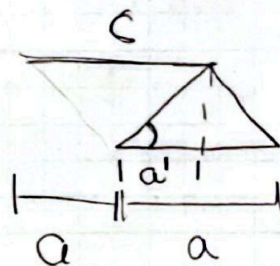


Joint C

FBD



angle at CE



$$\sqrt{.2^2 + .3^2} = h$$

$$h = 0.361$$

$$c - a = a'$$

$$= .2 \text{ m}$$

$$\sum F_y = -P + BC\left(\frac{b}{.5}\right) + CE\left(\frac{.3}{.361}\right) = 0$$

$$CE\left(\frac{.3}{.361}\right) = -BC\left(\frac{b}{.5}\right) + P$$

$$CE = (-BC\left(\frac{b}{.5}\right) + P)\left(\frac{.361}{.3}\right)$$

$$= (-45200\left(\frac{.3}{.5}\right) + 27.1 \times 10^3)\left(\frac{.361}{.3}\right)$$

$$CE = -24.1 \text{ N} = 24.1 \text{ NC}$$

$$\sum F_x = CD + CE\left(\frac{.2}{.361}\right) - BC\left(\frac{a}{.5}\right)$$

$$CD = -CE\left(\frac{.2}{.361}\right) + BC\left(\frac{a}{.5}\right)$$

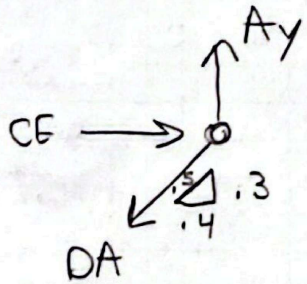
$$= +24.1\left(\frac{.2}{.361}\right) + 45200\left(\frac{.4}{.5}\right)$$

$$CD = 36200 \text{ N T}$$



Joint A

FBD



Given's

$$CE = 24.1N, A_y = 27100N$$

$$DA = 45267N$$

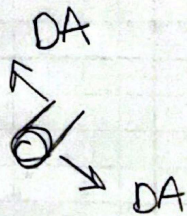


find: The crosssectional area of the connecting pin made of hardened tool steel

Givens:

$$S_y = 170 \text{ ksi}, \rho = 0.278 \text{ lb/in}^3, DA = 45300 \text{ N}$$

unknowns:  $A_{min}$ , weight of the pin



Convert to SI

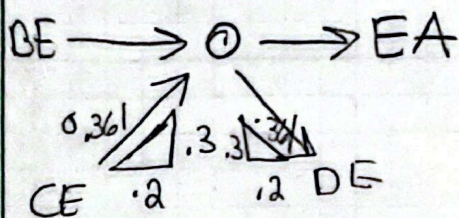
$$170 \text{ ksi} \times \frac{6.8948 \text{ MPa}}{1 \text{ ksi}} = 1172.1 \text{ MPa}$$

$$\frac{0.278 \text{ lb}}{\text{in}^3} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \times \left( \frac{1 \text{ in}}{0.0254 \text{ m}} \right)^3 = 7695 \text{ kg/m}^3$$



Joint E

FBD



Given:  $BE = 36200\text{ N}$

$CE = 24.1\text{ N}$

$$\sum F_y = CE\left(\frac{0.3}{0.361}\right) - DE\left(\frac{0.3}{0.361}\right)$$

$$CE\left(\frac{0.3}{0.361}\right) = DE\left(\frac{0.3}{0.361}\right)$$

$$CE = DE = 24.1\text{ N}$$

$$\sum F_x = +BE + EA + CE\left(\frac{0.2}{0.361}\right) + DE\left(\frac{0.2}{0.361}\right)$$

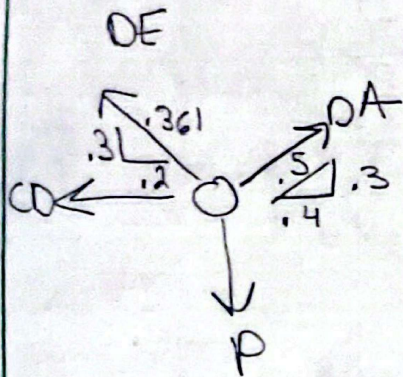
$$EA = -BE - CE\left(\frac{0.2}{0.361}\right) - DE\left(\frac{0.2}{0.361}\right)$$

$$EA = -36200 - 2.41\left(\frac{0.2}{0.361}\right)(2)$$

$$EA = -36203\text{ N} = 36203\text{ NC}$$



Joint D



Given's:

$$DE = 24.1 \text{ N}, CD = 36200 \text{ N}$$

$$P = 27.1 \text{ kN},$$

$$\sum F_x = DA \left( \frac{.4}{.5} \right) - CD - DE \left( \frac{.2}{.361} \right)$$

$$DA \left( \frac{.4}{.5} \right) = CD + DE \left( \frac{.2}{.361} \right)$$

$$DA = \left( CD + DE \left( \frac{.2}{.361} \right) \right) \left( \frac{.5}{.4} \right)$$

$$DA = \left( 36200 + 24.1 \left( \frac{.2}{.361} \right) \right) \left( \frac{.5}{.4} \right)$$

$$DA = 45267 \text{ N T}$$